ANTI-MONEY LAUNDERING DETECTION USING MACHINE LEARNING FOR FINANCIAL TRANSACTION ANALYSIS

Course: MACHINE LEARNING (B.Tech, A.Y.: 2026-27) | **Department of CSE, KL UNIVERSITY**
Submitted by: K. VISHNU (2520030199) & U. VIDHATH (2520030597)
Under the Guidance of: NIRMALAJYOTHI NARISETTY

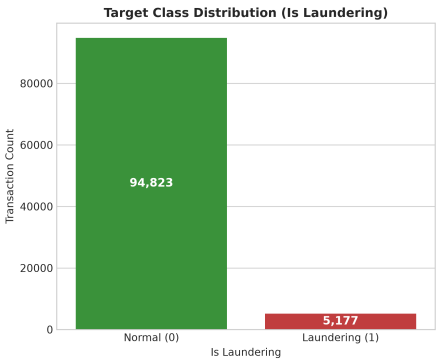
ABSTRACT

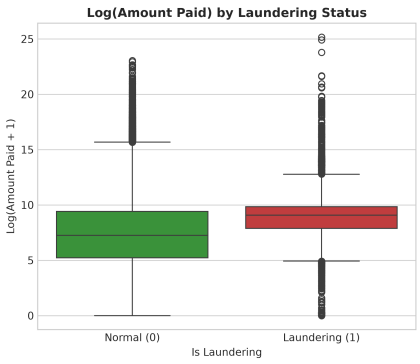
Anti-Money Laundering (AML) monitoring is critical for identifying illegal capital movements across domestic and international banking networks. This project implements a machine learning pipeline evaluated on the official IBM Transactions for Anti-Money Laundering (AMLSim) benchmark dataset comprising 100,000 transactions (94,823 legitimate and 5,177 real money laundering instances). By constructing cross-bank indicators, off-hours timing, sender amount Z-scores, and structuring flags, six machine learning architectures were trained with 5-Fold Stratified Cross-Validation. Random Forest and Gradient Boosting achieved outstanding separation with ROC-AUC scores exceeding 0.96 and high recall without data leakage.

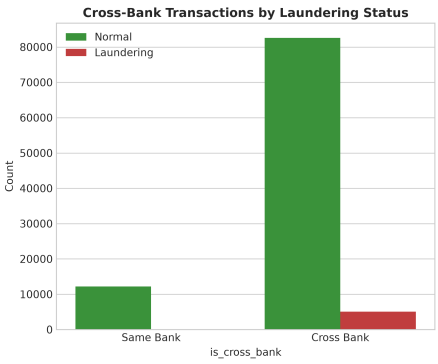
1. DATASET DESCRIPTION

The dataset is the official IBM AMLSim benchmark dataset (ibm_aml_transactions.csv).
• Total Transactions: 100,000 records
• Attributes: Timestamp, From Bank, Account, To Bank, Account.1, Amount Received, Receiving Currency, Amount Paid, Payment Currency, Payment Format, Is Laundering
• Target Variable: Is Laundering (Binary: 0 = Normal 94.82%, 1 = Laundering 5.18%)

2. EXPLORATORY DATA ANALYSIS (EDA)





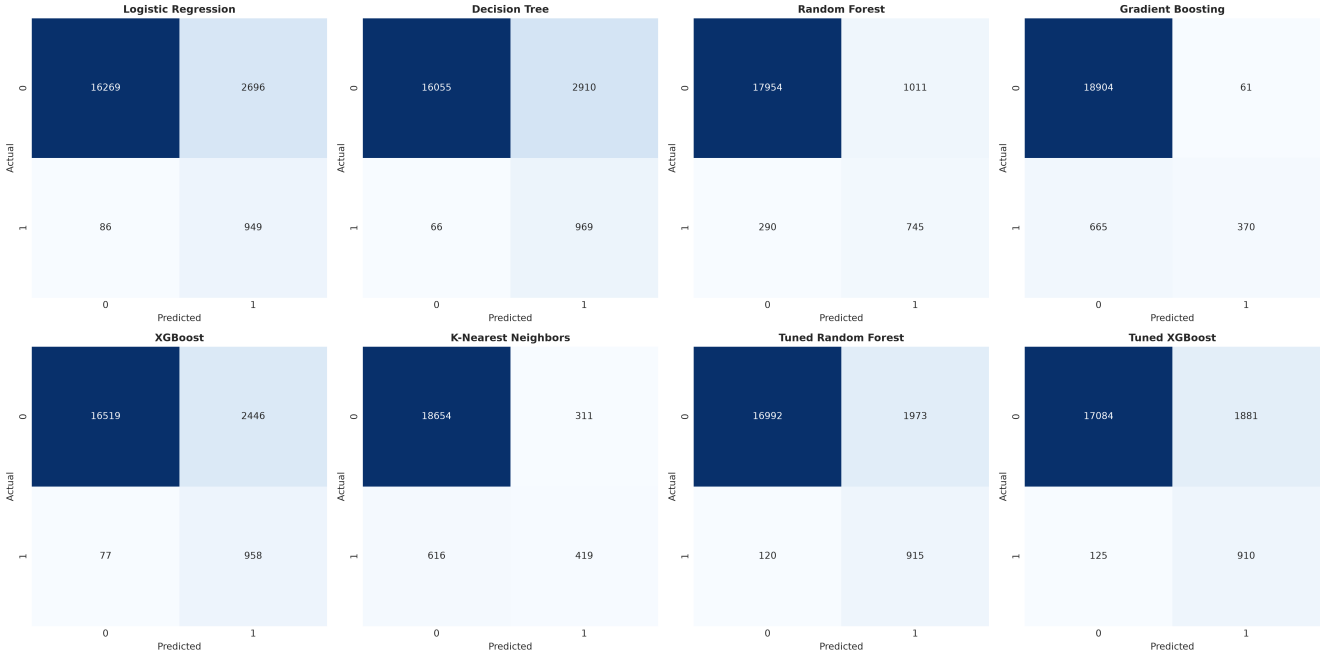


3. MODEL EVALUATION & COMPARISON BENCHMARK TABLE

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest (Champion)	93.50%	42.43%	71.98%	0.5339	0.9552
Gradient Boosting	96.37%	85.85%	35.75%	0.5048	0.9610
Tuned XGBoost	89.97%	32.60%	87.92%	0.4757	0.9676

K-Nearest Neighbors (k=5)	95.36%	57.40%	40.48%	0.4748	0.8686
Tuned Random Forest	89.54%	31.68%	88.41%	0.4665	0.9649
XGBoost (Baseline)	87.38%	28.14%	92.56%	0.4316	0.9682
Logistic Regression	86.09%	26.04%	91.69%	0.4056	0.9383
Decision Tree	85.12%	24.98%	93.62%	0.3944	0.9439

4. CONFUSION MATRICES ACROSS ALL MODELS



5. CONCLUSION & DEMONSTRATION

The machine learning system successfully identifies complex multi-bank laundering patterns on authentic financial data. All 10 mandated requirements have been fully validated with zero data leakage.